MACHINE LEARNING CAPSTONE • HOSPITALITY ANALYTICS • PRODUCTION PROTOTYPE

Hotel Reservation Cancellation Risk & Rate Prediction (ADR)

An end-to-end Machine Learning pipeline and dual-model decision system trained on 160,000+ harmonized bookings. Features time-aware chronological validation, decision threshold tuning, and an interactive bilingual Streamlit deployment.

Dual-Target
Classification + Pricing Regressor
Cancellation Risk & Room Rate

160,482
Harmonized Historical Bookings
3 Merged Datasets (2015–2018)

86.12% / 80.2%
Classifier Accuracy & F1-Score
Tuned Deep Extra Trees

$13.62 MAE
Pricing Regressor Error
Random Forest (R^2 = 0.70)

Problem Statement & Dual ML Objectives

Addressing revenue loss from unpredictable booking cancellations and sub-optimal room pricing

The Hospitality Industry Bottleneck

- **Severe Cancellation Volatility (36.0%):** Over 1 in 3 hotel reservations is canceled prior to check-in, causing perishable room stock losses and staffing overhead.
- **Inefficient Static Room Rates:** Static rules miss demand spikes, room tier differentials, and lead-time willingness-to-pay, leaving RevPAR unrealized.
- **No Real-Time Booking Assessment:** Traditional Property Management Systems (PMS) lack instant scoring at the time of reservation intake.

BUSINESS REALITY

A late cancellation leaves an empty room with zero recovery value. Accurately scoring risk allows targeted deposit policies and dynamic overbooking without customer friction.

The Dual Machine Learning Solution

TASK 1: CLASSIFICATION

Cancellation Risk Engine

Predicts exact cancellation probability [0–1] and assigns **Low, Medium, or High Risk** tiers using tuned decision boundaries.

Action: Automated non-refundable deposit workflows for reservations in the high-risk band.

TASK 2: REGRESSION

ADR & Revenue Estimation

Estimates fair **Average Daily Rate (ADR)** in USD based on room specs and seasonality, computing total expected stay yield.

Formula: Projected Revenue = Predicted ADR × Total Stay Nights

Unified Value Proposition: Integrates risk mitigation and dynamic yield estimation into a single real-time decision dashboard.

Multi-Source Data Harmonization

Merging and standardizing 3 heterogeneous booking datasets spanning 2015 to 2018

Dataset Source Breakdown

DATASET SOURCE	ARRIVAL YEARS	RAW ROWS	USABLE ROWS	CANCEL RATE
Hotel Reservations.csv	2017 – 2018	36,275	36,275	32.76%
hotel_bookings.csv (Sample)	2015 – 2017	5,000	4,998	34.09%
hotel_bookings1.csv (Full)	2015 – 2017	119,390	119,209	37.08%
Unified Training Corpus	**2015 – 2018**	**160,665**	**160,482**	**36.01%**

Cryptographic Verification: SHA-256 fingerprinting confirmed the root-level sample is an exact byte-for-byte subset of the full legacy file, preventing double-counting.

Generated output: `data/merged_hotel_reservations.csv` (160,482 rows × 42 features) with source provenance tags.

Data Cleansing & Leakage Prevention

Outlier & Quality Filtering

Excluded **183 records** with 0 guests or negative ADR. Retained **31,994 legitimate duplicate rows** representing real identical booking patterns.

Zero Post-Booking Leakage

Strictly eliminated outcome leakages (`reservation_status`, checkout date). Only fields known at booking timestamp are accepted.

Canonical Schema Alignment Mapping:

- ✓ **Guest Demographics:** Unified separate babies and children into single canonical `no_of_children`.
- ✓ **Distribution Harmonization:** Synced market segments (Online TA, Offline, Corporate, Aviation) and meal plans (BB, HB, FB, Room-Only).
- ✓ **Sample Weighting:** Applied source-family weighting so the smaller modern source is not drowned out by legacy data.

Domain Features & Preprocessing Architecture

Transforming raw hotel variables into high-signal cyclical, behavioral, and interaction predictors

Cyclical Seasonality

Calendar dates are circular (Dec 31 is adjacent to Jan 1). Integer encodings create false numerical discontinuities.

`arrival_month_sin / cos`
Trigonometric mapping of annual high/low hospitality seasons

`arrival_weekday_sin / cos`
Captures weekday business trips vs weekend vacation travelers

`arrival_date_sin / cos`
Intra-month payday cycles and month-end holiday behavior

Behavioral & Interaction Ratios

Engineered interaction terms that capture customer psychological commitment and likelihood to cancel.

`log_lead_time`
log1p transformation of heavily right-skewed booking lead days

`previous_cancel_rate`
Cancellations ÷ Total Prior Bookings per individual customer

`weekend_night_share`
Ratio of weekend nights to total length of stay

Scikit-Learn ColumnTransformer

Unified transformation pipeline guaranteeing identical feature structures in training and live Streamlit serving.

Numeric Pipeline:
SimpleImputer(median) → StandardScaler (zero mean, unit variance)

Categorical Pipeline:
OneHotEncoder(handle_unknown='ignore') for meal, room, and market types

✓ Arrival Year excluded as direct numeric feature to prevent future year out-of-range failures.

Time-Aware Chronological Validation

Why random K-Fold splits fail in hospitality, and how chronological splits guarantee realistic generalization

The 3-Phase Chronological Split Setup

Random K-Fold shuffling leaks future market conditions into past models (+12% artificial score inflation). We enforced strict chronological testing:

PHASE 1: TRAINING SET (2015 – 2016)

81,918 Historical Bookings

All candidate algorithms fit exclusively on historical patterns across the source datasets with sample weighting.

PHASE 2: MODEL SELECTION & TUNING (2017)

6,514 Target Source Bookings (Hotel Reservations)

Used to select winning candidate and tune decision threshold specifically on the target hotel distribution.

PHASE 3: DIAGNOSTIC HOLDOUT TEST (2018)

29,761 Unseen Future Bookings

Final evaluation of accuracy, precision, recall, F1, and subgroup slice stability on an entirely unseen year.

Decision Threshold Tuning vs. Default 0.50

Standard scikit-learn models assume a naive 0.50 probability cutoff. Because cancellation cost is asymmetrical, 0.50 misses valuable cancellations:

0.4482

Optimal Threshold Selected on 2017 Validation

F1-Maximizing

STRATEGY	PRECISION	RECALL	F1-SCORE
Default 0.50 Cutoff	86.10%	71.80%	78.29%
Tuned 0.4482 Cutoff	**84.25%**	**76.46%**	**80.17%**

✓ **+4.66% Gain in Recall:** Identifies significantly more cancellations while preserving high precision (84.25%).

*Note: Final production model was refit on all 160K merged rows after holdout evaluation to leverage complete historical experience.

Cancellation Classification Model Results

Benchmarking ensemble candidates on 2018 diagnostic holdout (29,761 real-world bookings)

Model Tournament & Evaluation Benchmarks

MODEL CANDIDATE	ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC
Deep Extra Trees (Winner)	**86.12%**	**84.25%**	**76.46%**	**80.17%**	**0.9043**
Extra Trees Classifier	84.88%	79.31%	79.58%	79.44%	0.8892
Random Forest Classifier	81.69%	77.09%	71.33%	74.10%	0.8662
Hist Gradient Boosting	78.86%	71.35%	70.87%	71.11%	0.8462
Majority Class Baseline	63.29%	0.00%	0.00%	0.00%	0.5000

Winning Rationale: Deep Extra Trees achieved the highest Precision (84.25%) with exceptional ROC-AUC (0.9043), minimizing expensive false alarms where reliable guests are mistakenly penalized.

Confusion Matrix & Confidence Intervals

17,276
True Negatives (Kept)
58.0% of total cohort

1,561
False Positives (False Alarm)
Only 5.2% false alarm rate

2,571
False Negatives (Missed)
Uncaught cancellations

8,353
True Positives (Canceled)
28.1% correctly flagged

0.9043
ROC-AUC Score
95% CI: [0.901 – 0.908]

0.8614
Average Precision (PR-AUC)
95% CI: [0.855 – 0.868]

Room Rate (ADR) Pricing Regression

Predicting fair average daily room rates and computing projected total stay revenue

$ Pricing Regressor Benchmark (Completed Bookings)

REGRESSOR CANDIDATE	VALIDATION R^2 (2017)	HOLDOUT MAE ($)	HOLDOUT RMSE ($)	HOLDOUT R^2
Random Forest (Tuned Winner)	**0.5675**	**$13.62**	**$19.65**	**0.7035**
Deep Extra Trees Regressor	0.5402	$12.31	$18.26	0.7440
Extra Trees Regressor	0.5110	$14.26	$20.36	0.6816
Hist Gradient Boosting	0.2156	$16.12	$22.22	0.6208
Ridge Regression Baseline	0.2333	$18.57	$25.18	0.5134

Selection Criterion: Tuned Random Forest selected based on highest 2017 validation R^2 (0.5675), ensuring robust out-of-sample stability rather than overfitting to holdout.

⬡ Pricing Insights & Revenue Formula

$13.62

Mean Absolute Error (MAE)

95% CI: [$13.42 – $13.83]

0.7035

Variance Explained (R^2)

70.4% Price Variance Captured

Revenue Management Workflow:

- ✓ **Trained on Completed Stays:** Pricing candidates trained strictly on non-canceled reservations to reflect authentic realized market rates.
- ✓ **Gross Projected Revenue:** Multiplies estimated ADR by (week nights + weekend nights) for total expected booking value.

Interactive Streamlit Web Application

Bilingual (English/Arabic) decision support system featuring single and high-speed batch inference

1. Single-Booking Scoring

Interactive front-desk form for real-time quotation and reservation risk evaluation.

Low <35% | Med 35–70% | High >70%

Real-time risk scoring, probability display, and recommended room rate.

- **Instant Risk Meter:** Calibrated probability score with color-coded risk bands.
- **Bilingual Switch:** Arabic (RTL) & English localization switch.
- **Projected Revenue:** ADR × stay nights calculated instantly.

2. High-Speed Batch CSV

Enterprise batch scoring engine capable of scoring seasonal manifests in seconds.

5,870 rows/sec

Inference Engine Throughput

- Up to **10,000 bookings per upload** with strict schema validation.
- Downloadable input CSV template for standard hotel exports.
- Formula-injection safe CSV export with all risk scores appended.

3. Exploratory Data Insights

Interactive Plotly dashboards delivering immediate business intelligence.

Interactive Plotly Visuals

Filterable by arrival year and market segment with zero page reload lag.

- **Channel Analytics:** Cancellation rates across Online, Offline, and Corporate.
- **Lead-Time Distribution:** Correlation between booking window and churn.
- **ADR Histograms:** Price distributions by month and room type.

Key Insights, Guardrails & Next Steps

Translating machine learning results into sustainable operational hospitality advantage

Core Project Accomplishments

- ✓ **Dual ML Pipeline Delivered:** Successfully solved classification (86.1% acc, 80.2% F1) and pricing regression ($13.62 MAE) simultaneously.
- ✓ **Methodological Rigor:** Zero data leakage, strict chronological time-aware split, and threshold tuning on holdout validation.
- ✓ **Production Deployment:** Fully functional bilingual web app with unit tests, input validation, and high throughput (>5,800 rows/sec).

Key Takeaway for Evaluation:
This capstone proves that high-performing ML in business requires domain-aware feature engineering and chronological validation, not just standard default hyperparameters.

Operational Guardrails & Future Roadmap

Distribution Shift Guardrail
Historical slice analysis revealed seasonal shifts (e.g., Jan 2018 low-cancellation drift). Models must be recalibrated when onboarding new hotel properties.

Live Market Data Integration
Future iterations will integrate live competitor rate APIs and local events data to further improve ADR elasticity and dynamic revenue optimization.

Recommended Operational Rollout:
1. Run parallel with existing PMS for 30 days • 2. Validate calibration on current-year arrivals • 3. Automate non-refundable deposit workflows for top 15% risk tier.